$H \rightarrow WW$ + charm

Analysis status

Chirayu Gupta · VUB · August 2026

2022postEE · 26.7 fb^{-1} · expected UL 1034

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Dedicated decks: `ctag-sf-deck.pdf` (34 slides) · `negrw-deck.pdf` (57 slides)

1 · Analysis and selection

Why H+c

The Higgs coupling to **charm** is the least constrained of the accessible Yukawas. Associated production **H + c** probes it directly, without relying on the $H \to c\bar{c}$ decay.

production	$pp \to H + c\ (+X)$
decay	$H \to WW^* \to 2\ell 2\nu$
final state	**opposite-sign eμ** + MET + ≥1 c-tagged jet
dataset	2022postEE, **26.7 fb^{-1}**, ReReco `22Sep2023`

> The eμ requirement removes $Z \to ee/\mu\mu$ by construction, so the irreducible background is **real $t\bar{t}$** rather than Drell-Yan.

Reference: **AN-23-102**, the full Run 2 analysis at 138 fb^{-1}.

Signal and final state

H → WW → 2ℓ2ν, produced with a charm quark.

Final state: **opposite-sign eμ** + MET + **≥1 c-tagged jet**

object	selection
muons	tight ID + tight PF iso, $p_T > 10$, $\lvert\eta\rvert < 2.4$
electrons	wp80iso, $p_T > 10$, $\lvert\eta\rvert < 2.5$, $\Delta R(e,\mu) > 0.4$
ll pair	OS, $p_T > 20$ / 10, pair $p_T > 30$, $m_{ll} > 12$
jets	$p_T > 30$, $\lvert\eta\rvert < 2.4$, tight-lepveto ID, $\Delta R(j,\ell) > 0.4$
c-jets	$p_T > 20$, **PNet medium** CvL/CvB
MET	PuppiMET

eμ removes the Z→ee/μμ peak **by construction** — the irreducible background is **real $t\bar{t}$**, not Drell-Yan. Triggers: MuonEG, SingleMu, SingleEle.

Signal region composition

process	SR yield	share of SR
$t\bar{t}$	17,744	82.3%
st	1,543	7.2%
vjets	1,523	7.1%
diboson	609	2.8%
higgsbkg	132	0.6%
H+c signal	**0.26**	**0.001%**
total	**21,552**	

The signal is **0.26 events** against ~21,600 background — which is why MC statistics, rather than data statistics, drives the sensitivity.

2 · MVA-defined regions

Six classes, one network

setting	value
model	`SimpleMLP_MultiClass` (b-hive)
classes	`hplusc, higgsbkg, tt, st, diboson, vjets`
features	**26**, including the **11 c-tag one-hot categories**
training	30 epochs, batch 1024, lr 1e-3, loss weighting on
split	80/20, deterministic by NanoAOD `event` id

> The 11 one-hot categories mean the network sees the **full 2D CvL/CvB plane** — the same information the scale factors are binned in, not a binary pass/fail.

argmax defines the regions

The winning class picks the channel; the winning score is the discriminant.

region	events	dominant process
SR_hplusc	21,552	**tt 82.3%**
CR_higgsbkg	9,220	tt 86.4%
CR_tt	44,199	**tt 94.1%**
CR_st	20,135	tt 87.6%
CR_diboson	6,585	tt 72.4%
CR_vjets	9,214	**vjets 43.2%**

Regions are **orthogonal by construction** — no cut optimisation to tune or defend, and every event is used somewhere.

Four of five CRs are tt-dominated — by design

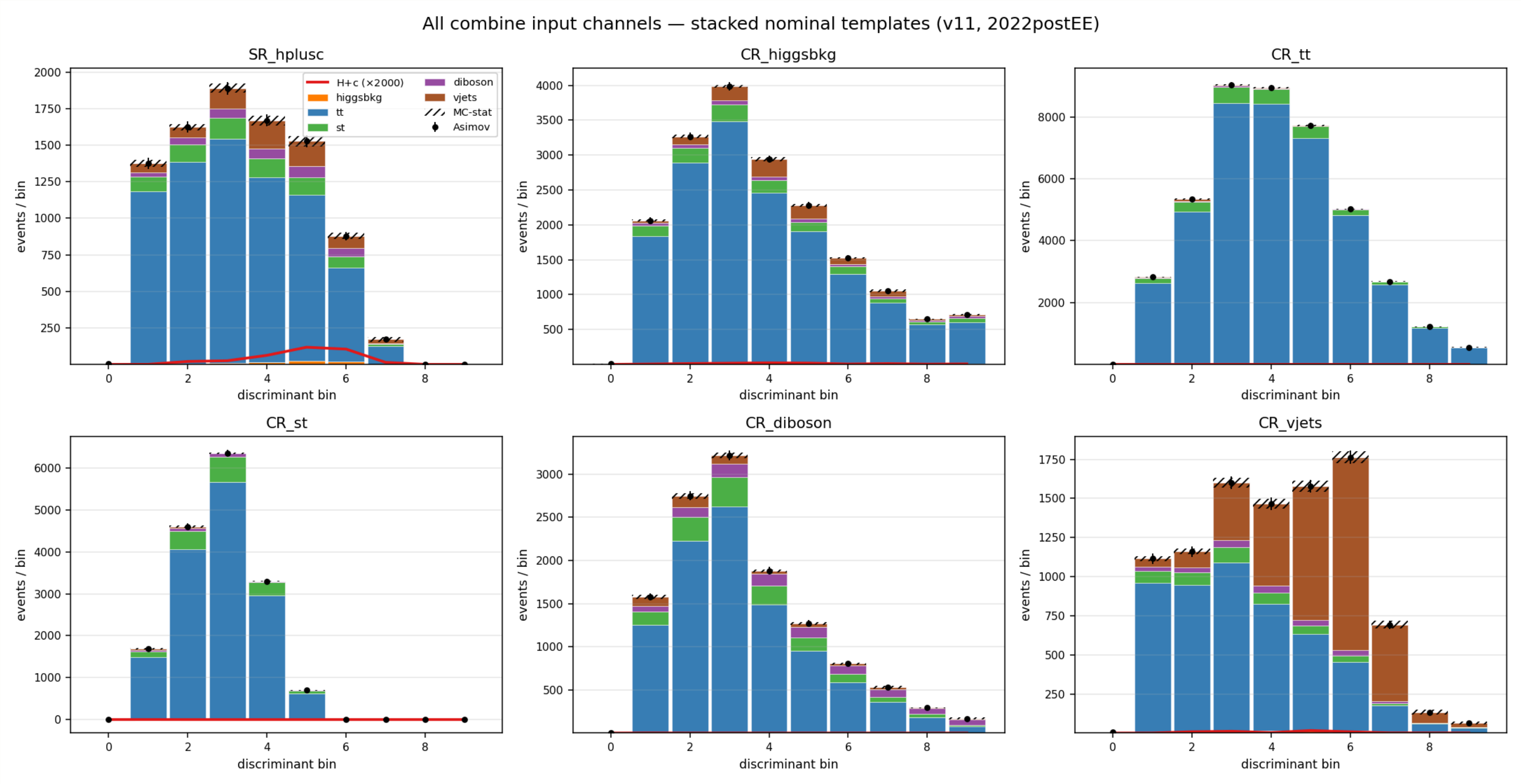
The SR is itself **82% tt**: in an eμ final state $t\bar{t}$ is dominant everywhere, so a tt-rich CR is expected, not a failure.

> **CR_tt: 94.1% purity over 44k events.** It pins the free-floating `rate_tt`, which covers 82% of the SR — the single most valuable constraint in the fit.

CR_vjets holds 59.6% of all V+jets in the fit. Without it there is no V+jets constraint from anywhere.

> Collapsing the CRs to single bins was **measured**: +33 units for three CRs, ~+50 for all five. The shapes carry real constraint, so all six channels keep 10 bins.

Templates entering the fit



6 channels × 6 processes × 10 bins. **CR_vjets** (bottom right) is the only region where V+jets (orange) is visible — it carries 59.6% of all V+jets in the fit.

3 · c-tagging

From working points to a plane

PNet gives **two** discriminants per jet: **CvL** (charm vs light) and **CvB** (charm vs bottom). A single working-point cut throws most of that away.

The calibration instead partitions the **whole plane** into **11 categories** `L0, C0–C4, B0–B4` — including the untagged `L0` bin.

property	value
axes	CvL, CvB — **verified sufficient**, no third variable needed
categories	11: `L0`, `C0`–`C4`, `B0`–`B4`
index	computed from CvL/CvB **already stored per jet**
applied	natively in the processor (`CTag2DCorrector`)

> Because the scheme spans the **full plane including `L0`**, the SF machinery works even for a selection with no working point at all.

Only 7 of 11 categories are populated

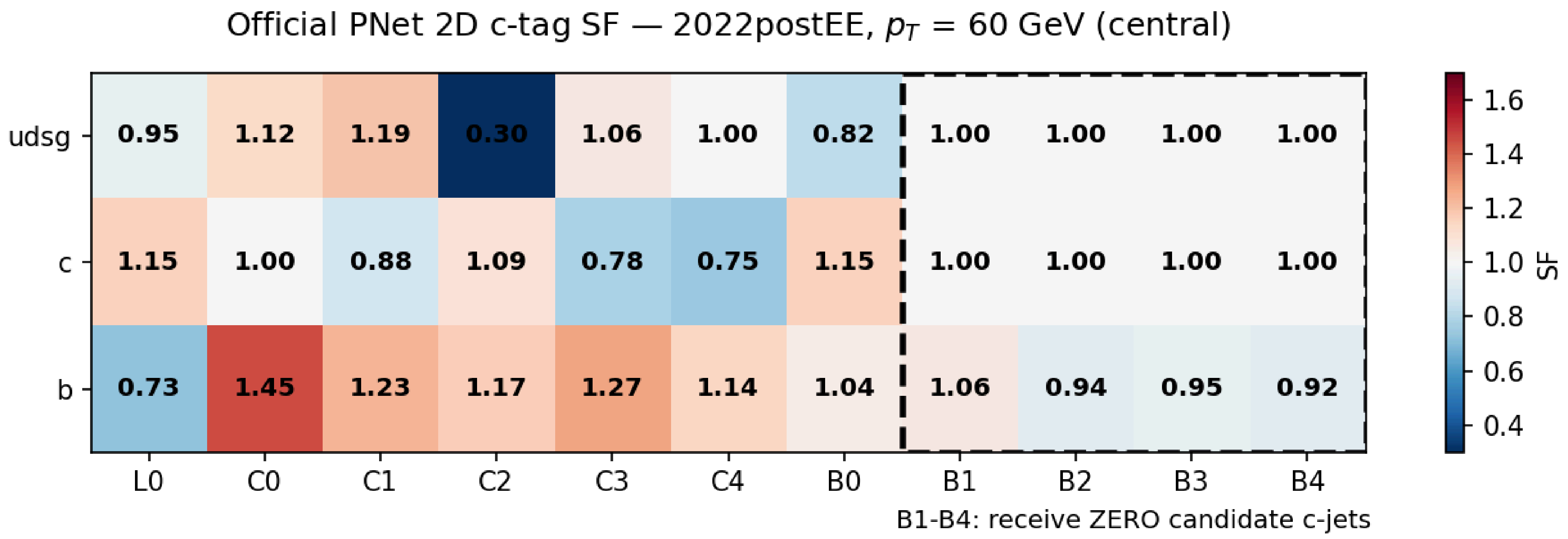
Occupancy of the five largest (`C4` and `B0` are populated but sparse):

category	jets	c (%)	b (%)	light (%)
`L0`	130,694	13.5	4.7	**81.8**
`C0`	188,702	28.2	6.2	65.7
`C1`	170,043	58.0	7.9	34.1
`C2`	17,823	58.6	33.4	8.0
`C3`	1,082	30.6	**57.9**	11.5

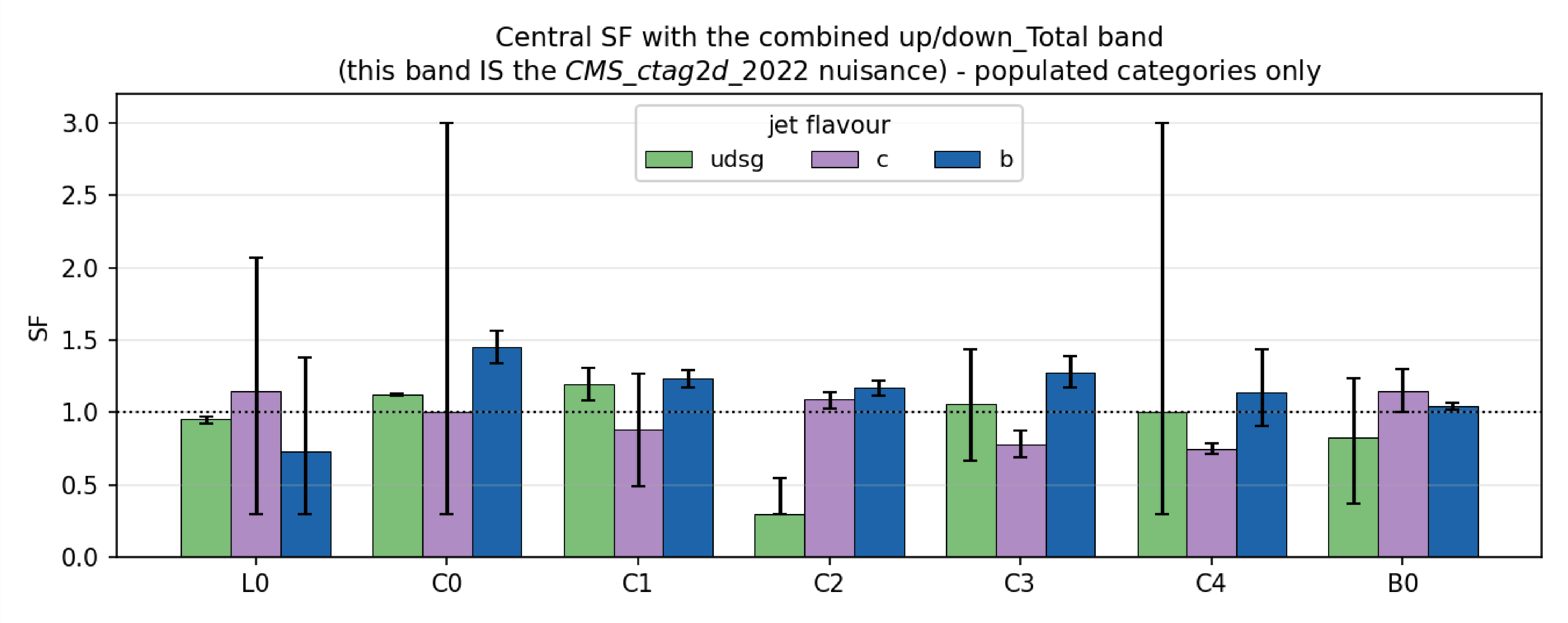
The scheme targets a broader phase space than ours — the b-rich `B1`–`B4` categories are empty after the eμ + c-jet selection.

> Category index is computed from CvL/CvB already stored per jet — **no NanoAOD reprocessing was ever needed**.

The scale factor matrix



The uncertainty band is the nuisance



What the calibration costs

configuration	limit
no SF	1150
+ `CMS_ctag2d_2022`	**1164**

Applying the calibration **worsens** the limit by 14 units — as it must.
A scale factor adds a nuisance. The point is correctness.

Currently **one nuisance** covering the whole plane. Decorrelation is pending.

4 · Negative-weight reweighting

full detail: `negrw-deck.pdf` *(57 slides)*

The estimator is a cancellation

aMC@NLO events carry **signed** weights. The yield estimator

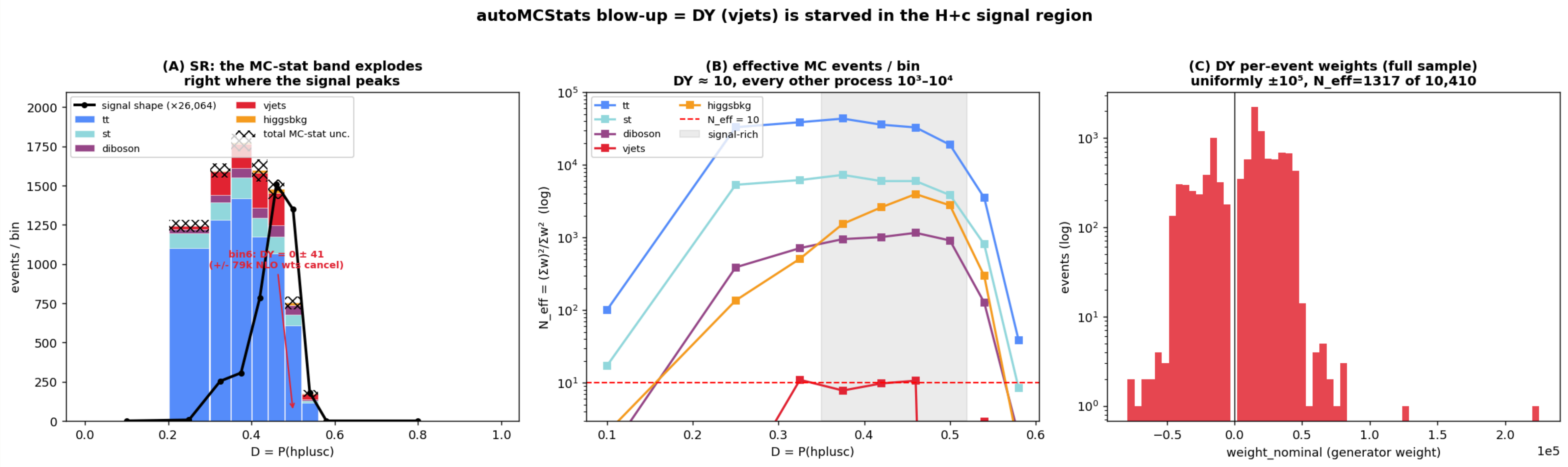
$$\hat{N}_B = \sum_{i \in B} w_i, \qquad w_i = \pm|w_i|$$

is a **difference of two large numbers**.

- the **expectation** is correct — the yield is unbiased
- the **variance** is not: it scales with $\sum w_i^2 = \sum |w_i|^2$, the *unsigned* statistics, while the mean is the much smaller signed sum
- effective statistics collapse: $N_{eff} = (\sum w)^2 / \sum w^2 \ll N$

16.4% of our V+jets events carried a negative weight. In a sparse bin the positives and negatives can cancel **completely** — leaving zero content with a large finite error.

V+jets was starved exactly under the signal



(A) the MC-stat band explodes where the signal peaks — one bin was DY $= 0 \pm 41$ from ±79k weights cancelling · **(B)** N_{eff} per bin: V+jets ≈10, every other process 10^3–10^4 · **(C)** DY per-event weights, uniformly $\pm 10^5$

The method — an algebraic identity

Write the signed generator density as a difference of two positive densities:

$$\mathrm{PDF}(\vec{x}) = a\,\mathrm{PDF}_+ - b\,\mathrm{PDF}_-, \qquad a - b = 1$$

With $P_+(\vec{x}) = a\mathrm{PDF}_+/(a\mathrm{PDF}_+ + b\mathrm{PDF}_-)$:

$$\mathrm{PDF}(\vec{x}) = \underbrace{(2P_+(\vec{x}) - 1)}_{g(\vec{x})} \cdot (a\mathrm{PDF}_+ + b\mathrm{PDF}_-)$$

The right factor is the **unsigned** density — what $\sum |w|$ samples.

> So $\sum |w|\, g$ and $\sum w$ estimate **the same distribution**. Yield-preserving by construction, not an approximation.

Exact for the *true* P_+. We use $\hat{P}_+$ from a finite classifier — so closure becomes an **empirical** property that must be measured.

Training region — train loose, infer tight

$g(\vec{x})$ is a **generator property**; it does not know about the analysis selection.
So we train on a far larger sample than the SR.

	region
train	all base cuts + veto of the eμ SR topology
infer	the tight eμ signal region

> **Disjoint by construction.** An event that is not "exactly one eμ pair" can never be an SR event — no event-id bookkeeping needed.

Rejected alternatives: inverting MET (kinematically softer → extrapolation) and inverting the c-jet requirement (flips into the *opposite* gen corner).

Inputs — 20 generator-level features

group	variables
LHE event	`lhe_njets`, `lhe_nb`, `lhe_nc`, `lhe_nuds`, `lhe_nglu`, `lhe_npnlo`
LHE kinematics	`lhe_ht`, `lhe_htincoming`, `lhe_vpt`, `lhe_alphas`
gen partons	multiplicity, `n_pt20/100/200`, incoming PDG IDs
leading partons	`genparton1/2_pt`, `genparton1/2_eta`

> **All generator-level.** The paper explicitly **rejects reco-level variables** — closure is only guaranteed on quantities the generator itself sampled.

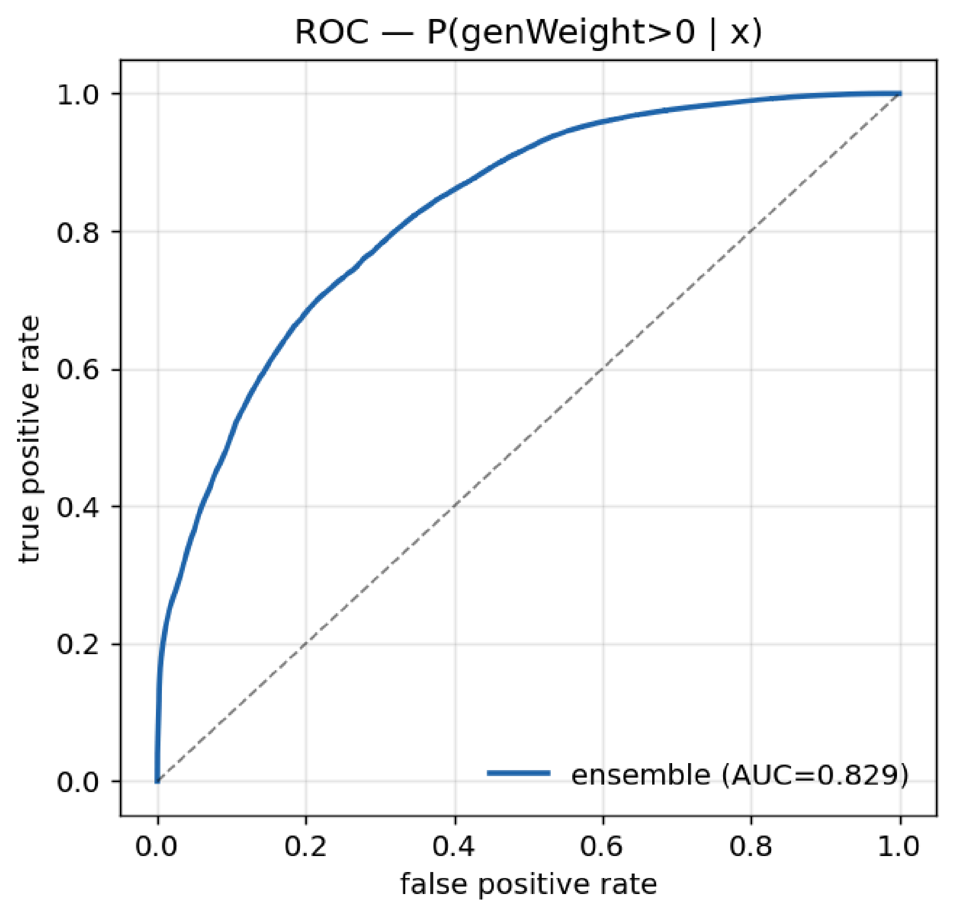
The two weight classes separate visibly in nearly every one of these features.

The classifier

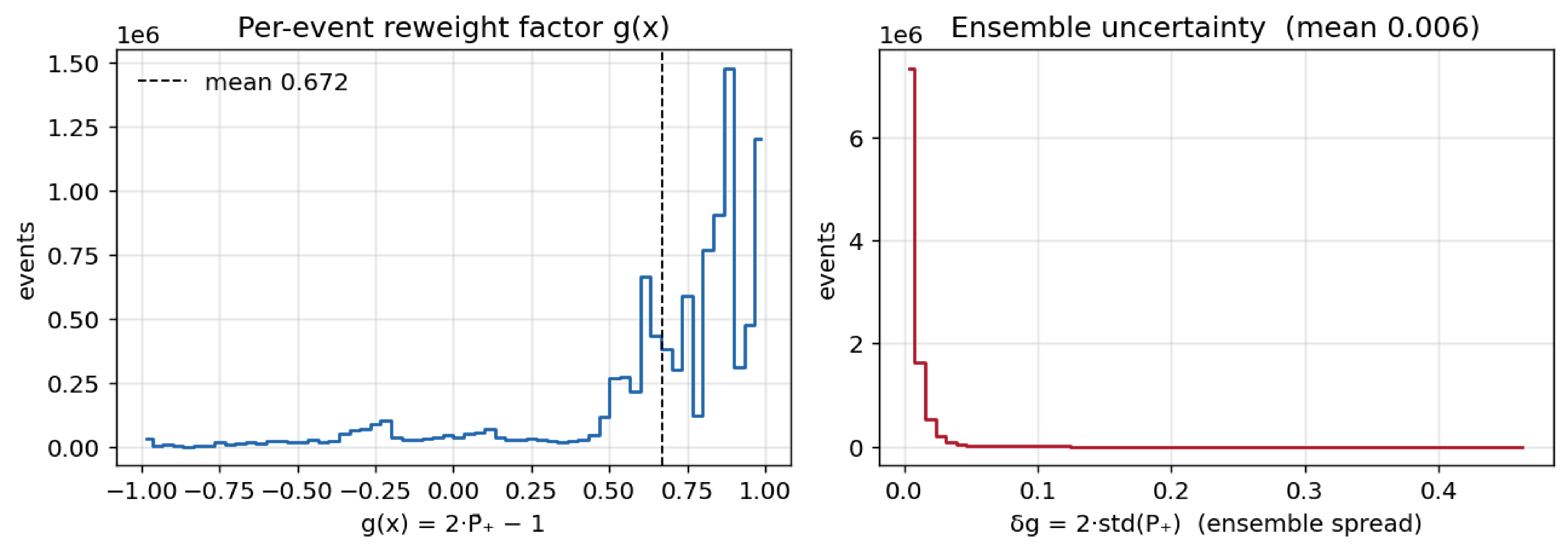
setting	value
model	`HistGradientBoostingClassifier` (scikit-learn)
inputs	20 generator-level (`lhe_*` , `genparton_*`)
training events	9.8M
ensemble AUC	**0.829**

AUC 0.83 on an **intrinsically stochastic** target — the weight sign is not a deterministic function of the kinematics, so a perfect classifier cannot exist even in principle. 0.83 means the generator-level features genuinely carry the information the NLO subtraction encodes.

Classifier ROC

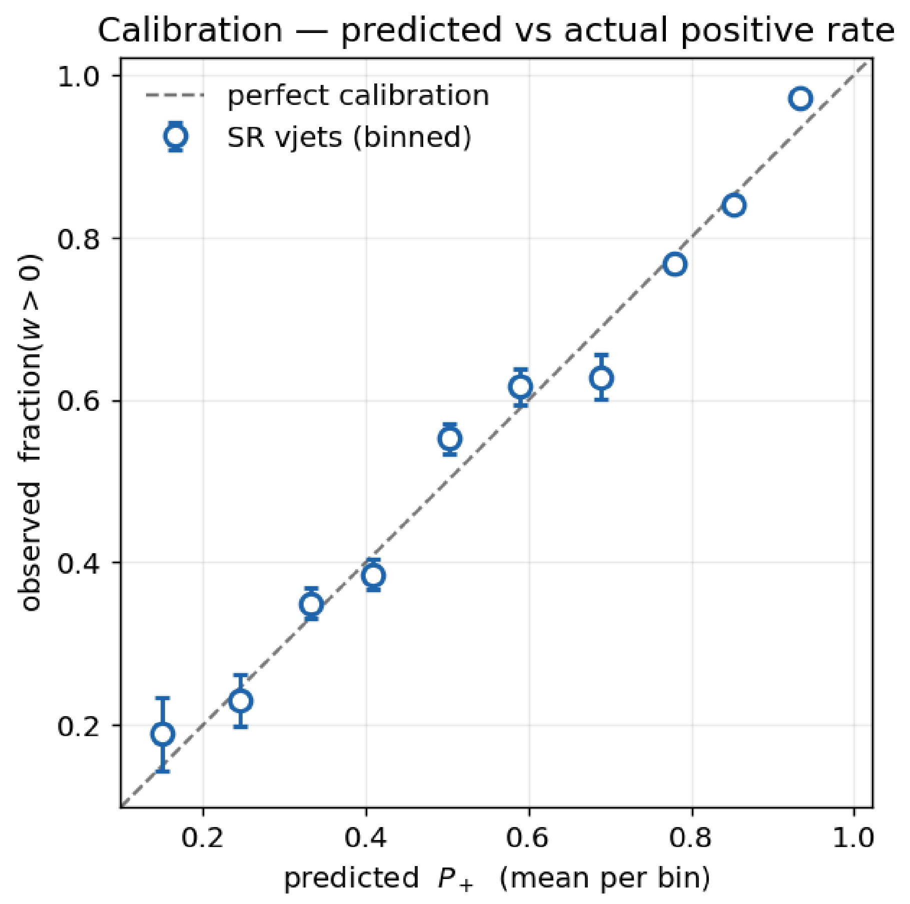


The reweight factor and its spread



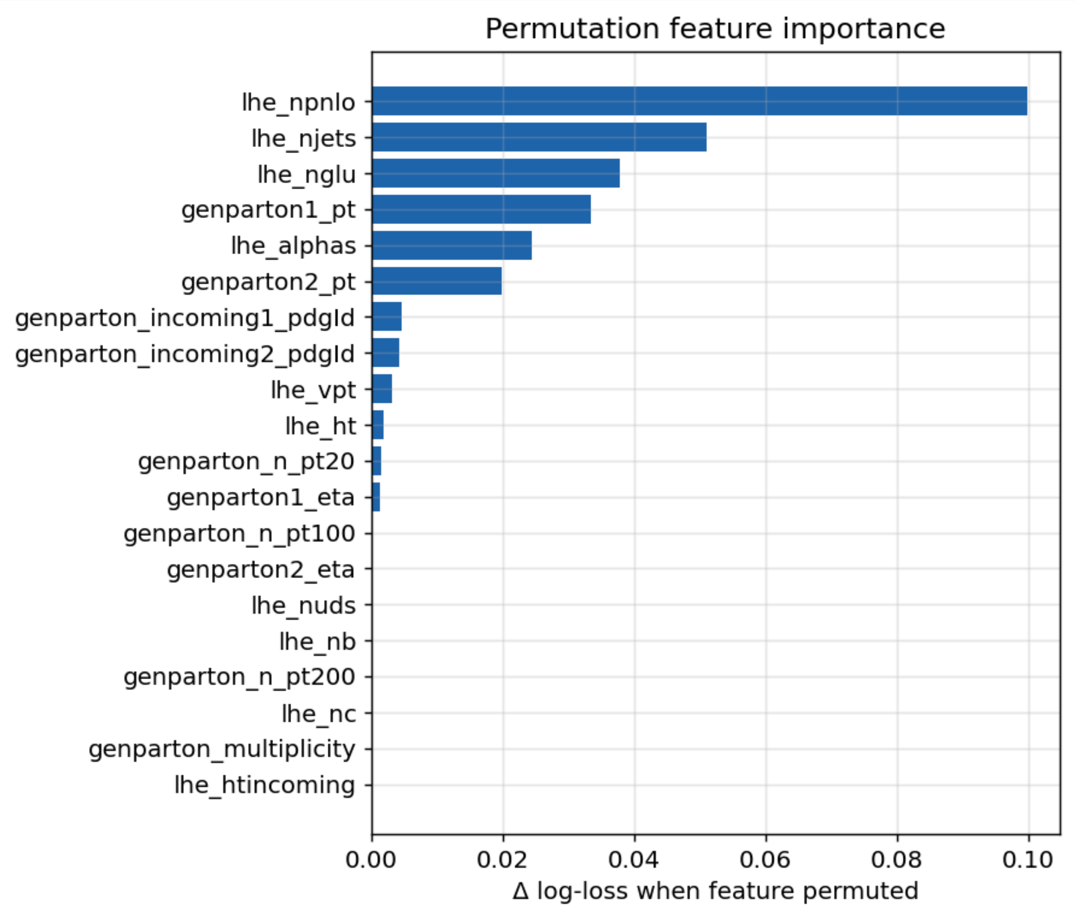
quantity	value	reading
g mean	**0.672**	$= 2 \times 0.836 - 1$, matching the global positive fraction
g range	$[-0.991,\ 0.993]$	inside the physical $[-1, 1]$ — no pathological events
δg mean / max	**0.006** / 0.467	20-model ensemble agreement is tight

Is P_+ calibrated outside the training region?



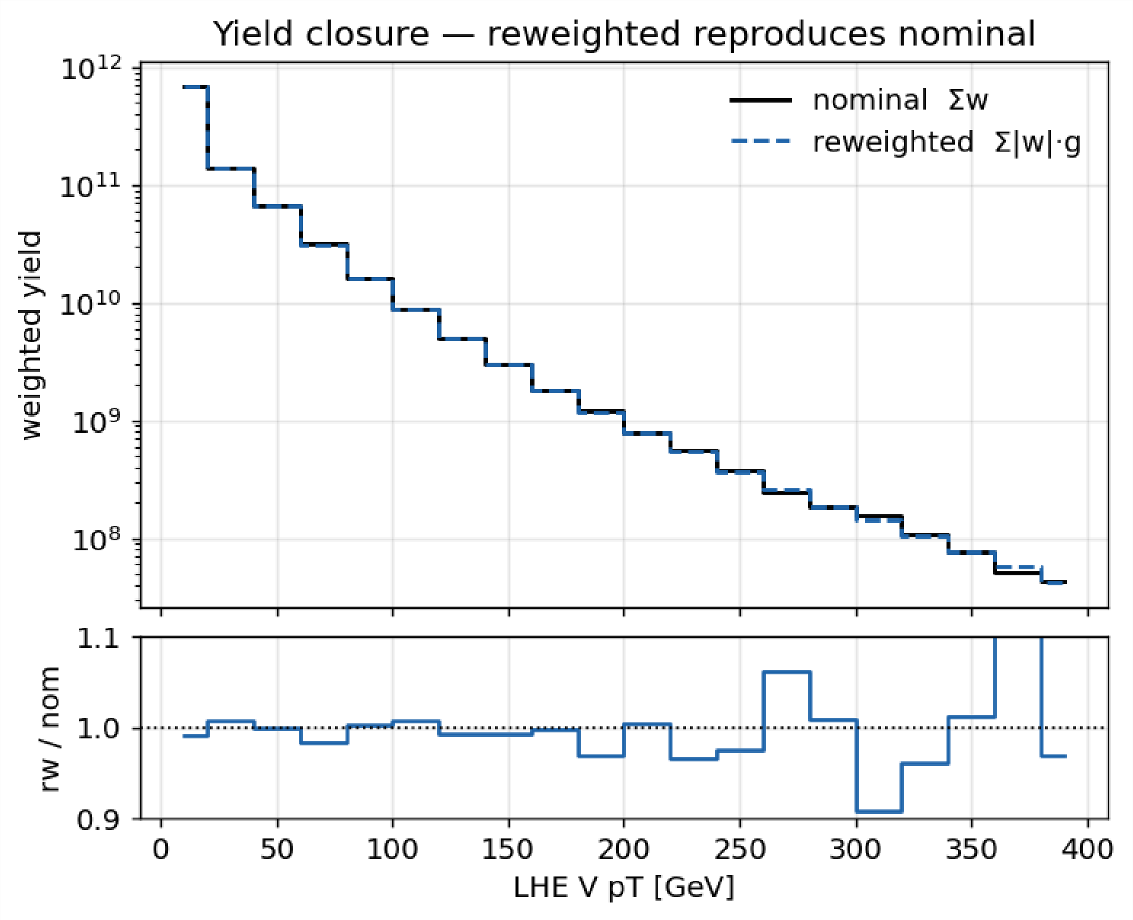
Bin SR events by **predicted** P_+, plot the **observed** fraction with $w > 0$.
Points land on the diagonal across the full range ($P_+ \approx 0.15 \to 0.93$) — the classifier is calibrated where it is applied, not only where it was trained.

Which features carry the information



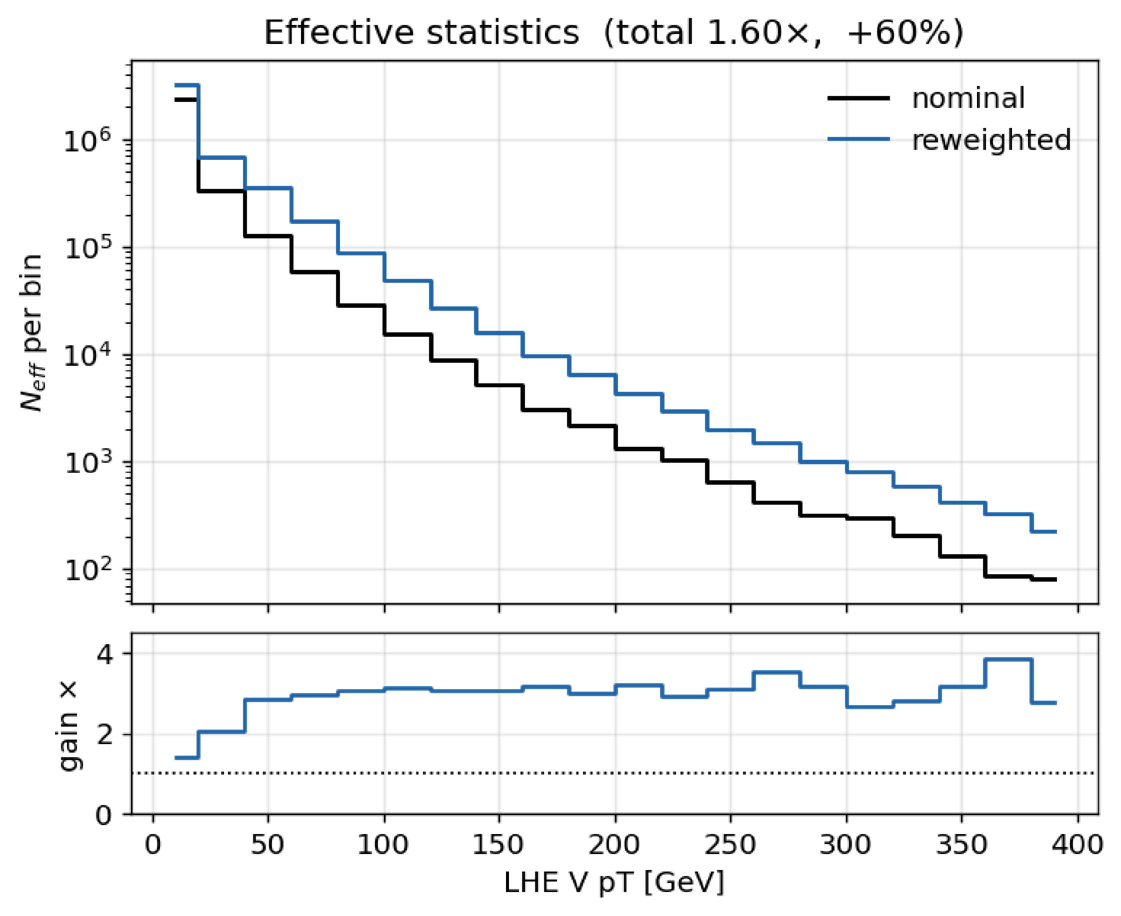
Permutation importance: the increase in log-loss when a feature is scrambled.

Closure



Training-region closure **0.994** on 9.8M events — the reweighting removes variance and nothing else.

The gain



Integrated gain **1.6×**; per-bin ratio **≈3×** in the signal-rich bins.

SR closure and the renormalisation

The SR is a small, kinematically-biased corner of the training domain, so the per-event g does not average to the *local* positive fraction. Applied **per dataset** in the card builder:

process	$\sum w$	$\sum \lvert w \rvert g$	renorm
DY (both mass bins)	1.014e8	1.027e8	**0.987**
W→ℓν	5.238e7	5.876e7	**0.891**

Applied **per dataset** in the card builder: the reweighted template is scaled back to the nominal yield, so the **yield is restored exactly** while the variance reduction — from per-bin spread, not the integral — is untouched.

These factors predate the W+jets sample change — they are for the old inclusive `WtoLNu_2Jets`. **The classifier will be retrained on the jet-binned 0J/1J/2J samples** and the factors re-derived.

5 · W+jets jet-binned samples

Inclusive → jet-binned

AN-23-102 §2.3 rejects the inclusive NLO sample outright:

"not used… 5 times smaller than LO and with large fraction of negative weights."

sample	xsec (pb)	events	files	neg-w
`WtoLNu-2Jets_0J`	55,760	678M	3,432	10.2%
`WtoLNu-2Jets_1J`	9,529	523M	2,669	25.7%
`WtoLNu-2Jets_2J`	3,532	345M	2,135	34.7%
sum	**68,821**	**1,546M**	**8,236**	
old inclusive	*67,710*	*282M*	*381*	*16.1%*

Cross sections from **XSDB**; sum within **+1.6%** of the inclusive — normal NLO merging spread, not a double-count. Measured negative-weight fractions match XSDB to ~1%.

> XSDB labels these `accuracy: "LO"` — **wrong**, auto-populated. `amcatnloFXFX` is NLO, proven by the 10–35% negative weights, impossible at LO.

Why the gain is not simply 5.5×

Raw events rise **5.5×**, but the useful quantity is effective statistics:

sample	n_{eff}/N	equivalent lumi
0J	0.635	7.7 /fb
1J	0.237	13.0 /fb
2J	0.094	9.1 /fb
combined		**29.8 /fb**
inclusive	*0.460*	***1.9 /fb***

The inclusive sample spends most of its cross section on **0-jet** events, which almost never survive the ≥1 c-jet requirement.

> Of the 4,851 surviving SR events, **2J alone contributes 69%** — despite being the smallest sample. The jet-binned samples are enriched in exactly what the selection needs.

W+jets result: 1160 → 1034

	before	after
expected UL	1160	**1034**
V+jets n_{eff} (SR)	280	**1170**
V+jets stat. error	5.98%	**2.92%**
V+jets rate	1508	1523

> **The rate barely moved while the statistical error halved** — the signature of a statistics gain, not a normalisation change.

6 · The combine card

Card structure

```
imax 6      # 6 channels: SR_hplusc + 5 argmax CRs
jmax 5      # 6 processes minus one
kmax *      # nuisances counted automatically

shapes * SR_hplusc  v11_hplusc_2dcat.root \
         SR_hplusc_$PROCESS  SR_hplusc_$PROCESS_$SYSTEMATIC
   ... one line per channel
```

channels	`SR_hplusc`, `CR_higgsbkg`, `CR_tt`, `CR_st`, `CR_diboson`, `CR_vjets`
processes	`hplusc`, `higgsbkg`, `tt`, `st`, `diboson`, `vjets`
binning	**10 bins per channel** — the winning MVA score
histograms	1,626 in the ROOT file (nominal + all up/down)
observation	`-1` in every channel — **Asimov, blind**

The two special entries

A free-floating tt normalisation:

```
rate_tt  rateParam  *  tt  1.0  [0,5]
```

One parameter, **shared across all six channels**, so CR_tt (94% pure, 44k events) determines the tt normalisation in the SR. This is why $t\bar{t}$ carries no `xsec` lnN — its rate comes from data, following AN-23-102.

Bin-by-bin MC statistics:

```
SR_hplusc    autoMCStats 10
CR_higgsbkg  autoMCStats 10
   ... one line per channel
```

Barlow–Beeston-lite. Threshold 10 means bins with $n_{eff} > 10$ get a single Gaussian nuisance; sparser bins get individual Poisson terms.

How a template becomes a card entry

```
parquet  →  read_scale = lumi × xsec / sumw  →  TH1 per (channel, process)
                                             →  + one TH1 per systematic ↑↓
```

Normalisation is a two-place mechanism. The per-event `genWeight` enters post-selection through the weights container; the denominator `sumw` is summed **pre-selection** and written to `sumw_records` — including chunks that select zero events. Reading it from parquet metadata instead undercounts.

Object shifts (JES/JER, lepton scale/res) cannot be weights — they change the MVA score, so each is a **separate parquet tree**, re-selected and re-scored, giving 12 shifted directories.

6 · Systematics

The full inventory

22 shape · 9 lnN · `rate_tt` free rateParam · autoMCStats (threshold 10)

shape (weight)	shape (object shift)
`pileup`	`CMS_scale_j_2022` (JES)
`ps_isr`, `ps_fsr`	`CMS_res_j_2022` (JER)
`scalevar_muR`, `_muF`, `_muR_muF`	`CMS_scale_e_2022`, `CMS_res_e_2022`
`lhe_pdf`, `lhe_alphaS`	`CMS_scale_m_2022`, `CMS_res_m_2022`
`muon_id`, `muon_iso`	
`electron_id`, `electron_reco` ×3	**lnN**
`CMS_ctag2d_2022`	`lumi_13p6TeV`, `xsec_st/diboson/vjets/higgsbkg`
`CMS_negrw_vjets`	`BR_HtoWW`, `BR_Htautau`, `xsec_hplusc_4FS_5FS`
`flavor_composition_ggH` (placeholder)	

Object shifts need full reprocessing

JES/JER and lepton scale/resolution **change the MVA score**, so they cannot be applied as an event weight.

Each is a **separate parquet tree**, re-run through the full selection *and* re-scored by the network — 12 shifted directories.

> Inference coverage over every shift directory is **verified after each rebuild** — a partially-scored set would leave object-shift templates frozen at nominal.

Top-p_T reweighting

Implemented following `hh2bbww` (the framework behind AN-24-091), using the **theory-based (NNLO/NLO)** parameterisation with the Run 3 rescaling:

property	value
form	theory-based NNLO/NLO
Run 3 factor	× (0.991 + 7.5e-5·p_T)
nominal	reweighted
p_T cap	none — well-behaved at high p_T

```
sf_run2 = 0.103*exp(-0.0118*pT) - 0.000134*pT + 0.973
sf      = (0.991 + 0.000075*pT) * sf_run2
weight  = sqrt(prod(sf))        # over the two gen tops
down    = 1.0                   # "no correction" is the variation
```

The variation is **down = 1.0** ("no correction"), symmetric about the nominal.

Higgs heavy-flavour

ggH is only **13.1%** of the merged `higgsbkg` group (VBF 29.0%, ggZH 23.3%, ZH 21.0%, WH 9.1%), so a flat lnN on the group either over-penalises the other 87% or must be diluted to an average — and cannot produce a shape effect.

Implemented as a per-event weight instead, keyed on gen-jet flavour.

> **Why our Bkg-Higgs impact stays small even once activated.** Freezing it moves the limit by <1 unit, against **7.6%** in AN-23-102. Two causes: the per-event weight is **not yet active**, *and* `higgsbkg` is only **0.6% of our SR** — ours is tt-dominated, theirs is Higgs-background-dominated. It will rise, but **not to 7.6%**.

Replaced with a per-event weight, ported from HiggsDNA `Higgs_plus_HF_syst`:

```
num_HF_jets = ak.sum(genJets.hadronFlavour == 4, axis=-1)   # pT>25, |eta|<2.5
up   = where(num_HF_jets > 0, 1.5, 1.0)      # AN-23-102: 50% on ggH
down = where(num_HF_jets > 0, 0.5, 1.0)
```

> Keys on **gen-jet flavour**, so process grouping stops mattering — and it produces

MET unclustered energy

> **Implemented, not yet in the card** — it needs the shifted trees from a reprocessing pass, so it does not appear in the 22 shape nuisances above.

For Run 3 PuppiMET the shift is taken directly from the NanoAOD branches:

```
events.PuppiMET.ptUnclusteredUp / ptUnclusteredDown
events.PuppiMET.phiUnclusteredUp / phiUnclusteredDown
```

Matches HiggsDNA `MET_syst_Unclustered` — same branches, same up/down ordering.

What is implemented

systematic	implementation
`CMS_negrw_vjets`	ensemble spread of the negative-weight reweighting
`CMS_ctag2d_2022`	2D CvL/CvB SF, applied natively in the processor
`electron_reco` ×3	p_T-binned (<20, 20–75, >75 GeV)
`lhe_pdf`, `lhe_alphaS`	NNPDF replicas, MC2Hessian
`top_pt`	hh2bbww theory form — *pending activation*
`higgs_plus_c`	per-event HF weight — *pending activation*
MET unclustered	object shift — *pending activation*

The last three are implemented and verified but **not in the current card** — each needs its weight column or shifted tree from a reprocessing pass.

Deliberately excluded

item	decision
muon reco SF	**not applicable** — HiggsDNA exposes no reco key; hh2bbww registers `mu_id_sf` / `mu_iso_sf` and no `mu_reco_sf`. Two frameworks, same conclusion.
PU jet ID	absent from both frameworks for Run 3
UE / tune	requires **dedicated tune samples** — cannot be a weight
`hdamp`, `mtop`	sample-based tt modelling — **under consideration**, see next slide

$t\bar{t}$ modelling: `hdamp` and `mtop`

Two standard Run 3 $t\bar{t}$ modelling uncertainties, both **not currently in the card**:

term	what it is
`hdamp`	The Powheg damping parameter controlling the matching between the NLO matrix element and the parton shower. It sets how much hard radiation comes from the ME rather than the shower, so varying it changes the **jet multiplicity and jet p_T spectrum** of $t\bar{t}$.
`mtop`	The top-quark mass used in generation. Varying it (typically ±1 GeV) shifts the $t\bar{t}$ **kinematics and acceptance**.

Neither can be applied as an event weight — both require **dedicated alternative samples** generated with the varied parameter.

> **Undecided whether to add these.** They are standard in Run 3 $t\bar{t}$ analyses, but absent from AN-23-102 Table 16, and $t\bar{t}$ normalisation here is already taken from data via the free `rate_tt`. Adding them means generating and processing new samples.

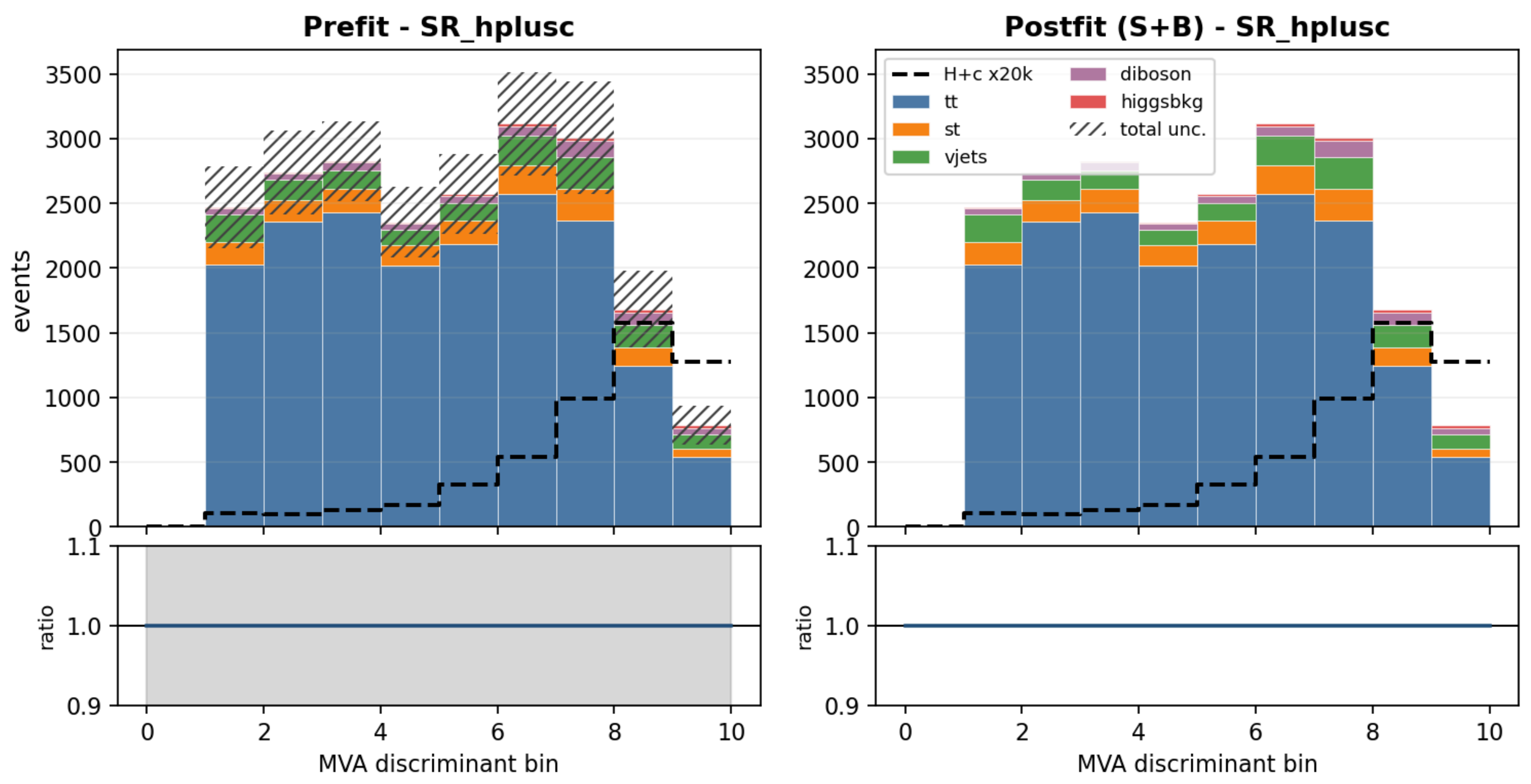
What moves the limit

Each nuisance frozen in turn; **Δ is the improvement in the expected limit.**

frozen	limit	Δ	% of 1034
all constrained (stat-only)	**641**	**393**	**38.0%**
theory shapes (scale + PS + α_S)	883	151	14.6%
`xsec_hplusc_4FS_5FS`	921	113	10.9%
`scalevar_muF`	944	90	8.7%
autoMCStats (all)	956	78	7.5%
`CMS_ctag2d_2022`	964	70	6.8%
`rate_tt`	1011	23	2.2%
JES	1023	11	1.1%

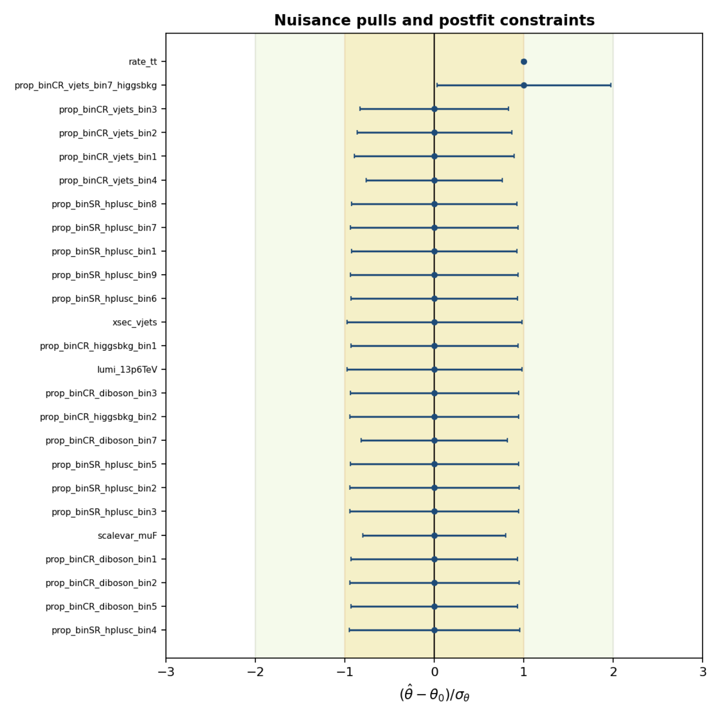
> **autoMCStats is no longer the leading term.** It was dominant before the W+jets change; at 7.5% the analysis is no longer MC-stat-limited in the way it was. Signal theory now leads.

Template composition — signal region



Backgrounds stacked, H+c overlaid (×20k). Hatched band = **prefit** total uncertainty, ≈14% per bin. Postfit errors are absent because combine writes a **zero postfit covariance** on an Asimov fit.

Nuisance pulls and constraints

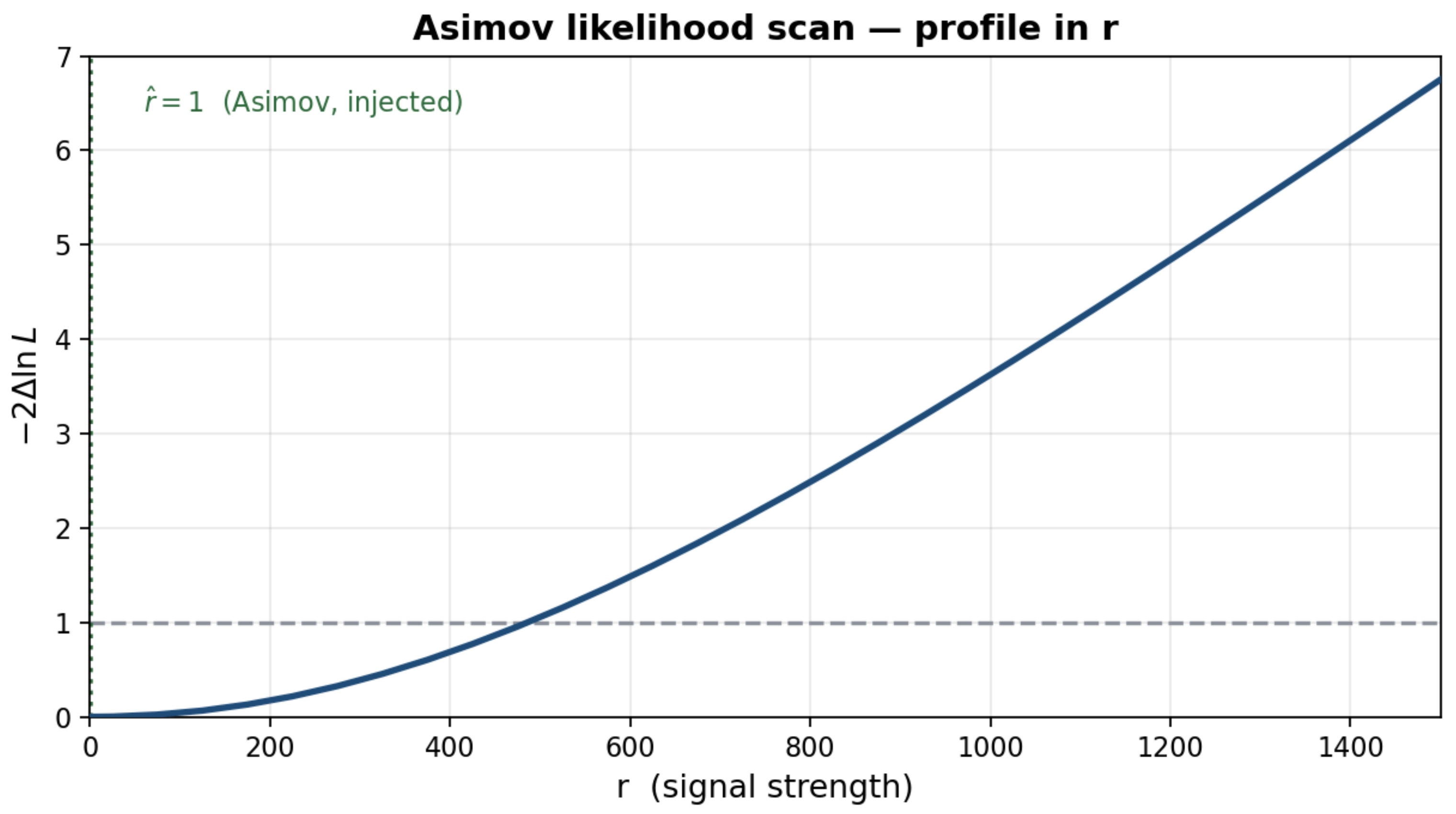


All Gaussian nuisances sit at **zero pull with ~unit width** — correct for Asimov.

The two non-zero entries are **not Gaussian** and are not pulls:

entry	value	why
`rate_tt`	1.000 ± 0.017	free rateParam, pinned by CR_tt

Likelihood scan



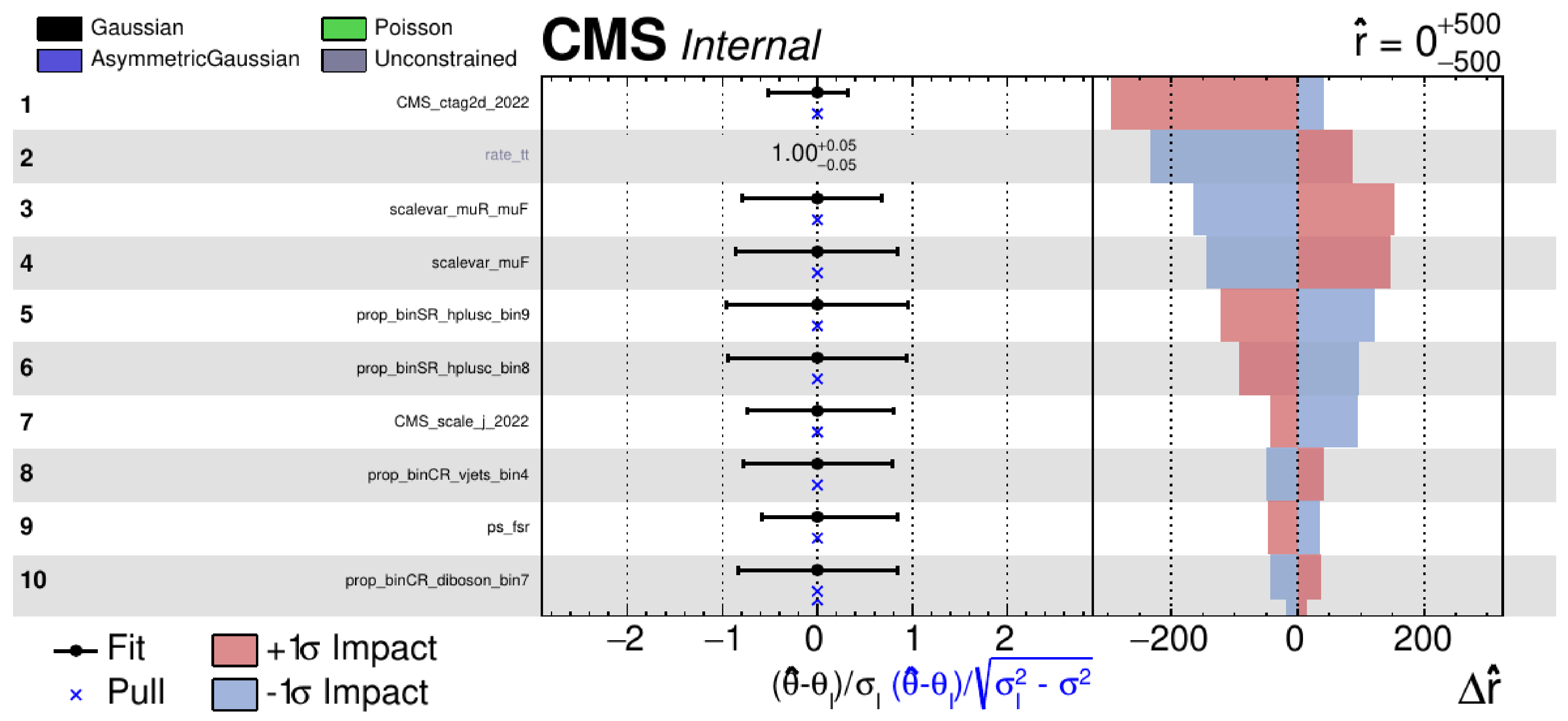
What the likelihood scan shows

The profile likelihood in the signal strength — all nuisances minimised at each point in r.

feature	value	meaning
minimum	**$\hat{r}$ = 1**	the fit recovers the injected Asimov signal — no bias
$-2\Delta \ln L = 1$	**r ≈ 525**	the 1σ interval on r
$-2\Delta \ln L = 3.84$	**r ≈ 1075**	the 2σ crossing
curvature	—	how fast the likelihood falls away = the sensitivity

> The 3.84 crossing (**1075**) sits close to the quoted CLs limit (**1034**) — the two constructions agree when the background-only and signal+background hypotheses are well separated, which is the regime here.

Nuisance impacts — top 10



The header `r̂ = 0 +500/−500` is a **plotImpacts rounding artefact** — the actual fit returns **$\hat{r}$ = 0.99996, −513/+483**, i.e. exactly the injected Asimov value.

Reading the impacts — the asymmetries

Left panel — the pull $(\hat{\theta} - \theta_0)/\sigma_\theta$. All pulls sit at zero: this is an **Asimov** fit, so by construction nothing is pulled. A non-zero pull here would mean the fit is being dragged, and would need explaining.

Right panel — $\Delta\hat{r}$ when each nuisance is moved by $\pm 1\sigma$.

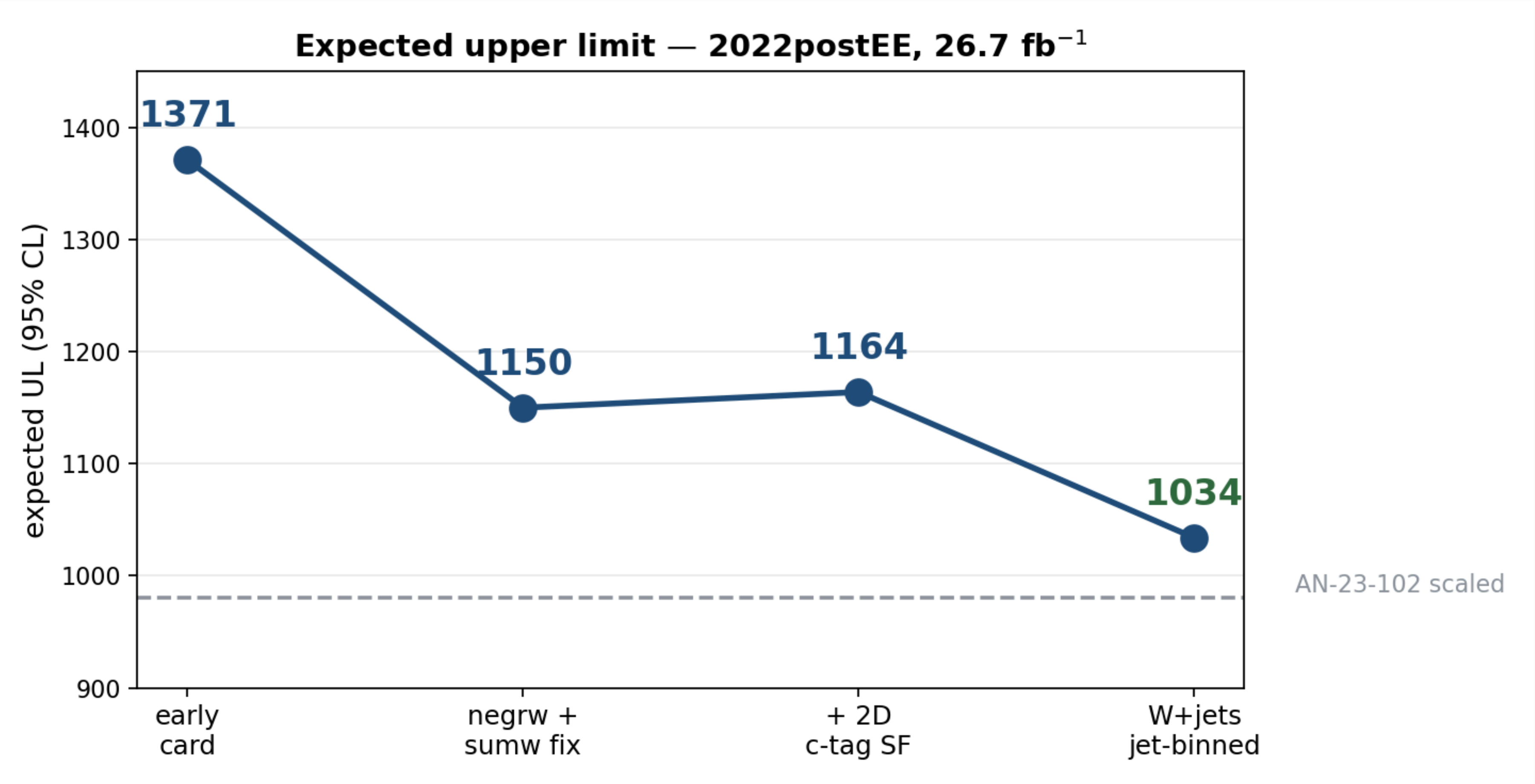
> **Red (+1σ) and blue (−1σ) are not mirror images.** Three sources of asymmetry:

source	why
asymmetric lnN	`xsec_st` is `0.9873/1.0167` — the up and down variations differ by construction
template shape	up/down shifts of JES or ctag move events between bins non-linearly
re-profiling	pushing one nuisance lets the others re-fit, and they respond differently in each direction

> The **fit range matters**. An earlier run with `--rMin 0` clipped every −1σ impact at the boundary and produced a one-sided plot. A symmetric range around $\hat{r}$ is required for the impacts to be meaningful.

7 · Status and outlook

The road so far



Note the suppressed zero on the y-axis — the c-tag step is a genuine but small **+14** increase, not the spike it appears to be.

Where we stand

	limit
AN-23-102 scaled to 26.7 fb^{-1}	**1144**
this analysis	**1034**
our statistical only	**641**

Better than the published analysis scaled to our luminosity — 1034 vs 1144, a **9.6% improvement**, on **5× less data**.

The scaling is indicative, not rigorous. $503 \times \sqrt{138/26.7} = 1144$ assumes a purely statistics-limited extrapolation. AN-23-102's Table 17 puts its statistical term at 73.8%, so it is *not* fully statistics-limited — the true scaled value would be somewhat **worse** than 1144, making our margin **larger** than 9.6%, not smaller.

The systematic breakdown of the 1034 is **under investigation** — the full term-by-term comparison against AN-23-102 Table 17 is not yet settled.

What is still missing

item	status
negrw retrain	classifier to be retrained on the jet-binned W+jets samples
4FS/5FS signal theory	placeholder 1.30 — needs re-derivation at 13.6 TeV
Trigger scale factors	implemented, **not yet enabled**
Higgs heavy-flavour (`higgs_plus_c`)	implemented, pending activation
top-p_T reweighting	implemented, pending activation
Dataset substitutions	W+jets p_T-binned (full AN stitch)
	DY → $Z\to\tau\tau$ filtered
	tt / single-top `-ext` samples
	`WZto3LNu`

Priorities

#	item	why
1	Re-derive `xsec_hplusc_4FS_5FS`	30% flat lnN on a statistics-starved signal
2	Enable trigger SFs	implemented, one flag
3	Activate `higgs_plus_c` + `top_pt`	proper HF and tt treatment
4	Add the missing datasets	more MC statistics
5	Decorrelate `CMS_ctag2d_2022`	currently one nuisance over the whole plane

Summary

Expected UL 1371 → 1034 · statistical-only **641**

V+jets n_{eff} **280 → 1170**

Within ~5% of the Run 2 analysis scaled to our luminosity

Remaining: 4FS/5FS re-derivation, trigger SFs, dataset substitutions